

# OOS Impact Analytics — Methodology Sheet

How the dashboard detects out-of-stock and demand-throttling events and values their revenue / contribution-margin impact. Worked example: **VV-VITA-311 — NAC Capsules** (amazon.it). Prepared for review.

## 1. Data sources

- **Amazon FBA Inventory Ledger** — real daily sellable warehouse balance and units shipped per SKU. The account runs **Pan-EU**, so stock is pooled: a SKU is physically out only when the whole EU balance hits zero.
- **Novadata margin export** — daily units, sales and contribution margin (CM3) per SKU and marketplace; supplies price and CM3 per unit for valuation.
- **Shopify catalog** — English product titles.

## 2. Two core measures

**Demand rate (lambda).** Trailing 90-day average units per calendar day, per SKU and marketplace. Daily sales follow an approximate Poisson process, so the chance of selling zero on a normal day is  $P(0)=e^{-\lambda}$ : ~37% at 1/day, ~5% at 3/day, ~0.7% at 5/day. A zero-sales day is therefore only treated as a stock-out when lambda is high enough that zero is a real anomaly (default threshold 3/day).

**Reach (days-of-supply).** EU sellable stock divided by trailing 28-day average units shipped — how many days of cover remain.

## 3. Daily classification (one label per SKU x marketplace x day)

Evaluated in priority order; the first match wins:

| Priority | Label           | Condition                                                                                                            | Counts as          |
|----------|-----------------|----------------------------------------------------------------------------------------------------------------------|--------------------|
| 1        | Physical OOS    | EU sellable balance = 0                                                                                              | Lost (involuntary) |
| 2        | Critically low  | Reach < 3 days                                                                                                       | Lost (involuntary) |
| 3        | Cooling down    | $3 \leq \text{reach} < 30$ days, still selling (units > 0), throttled (price $\geq +8\%$ and/or ad cut $\geq 50\%$ ) | Miss (voluntary)   |
| 4        | Marketplace gap | Units = 0 on a day enclosed by sales, $\lambda \geq 3/\text{day}$ (incl. throttles that hit 0)                       | Lost (involuntary) |

Guards: the SKU must already be selling (ignores pre-launch); a zero-run must be enclosed by later sales or be within the last 21 days (ignores discontinued tails). Cooling-down separates deliberate throttling — which otherwise looks like a demand drop — from genuine stock-outs, so the two are never double-counted.

## 4. Impact valuation

On any flagged day: **missed units =  $\max(\lambda - \text{actual units}, 0)$**  (the expected demand we didn't capture). Valued at the SKU's trailing average selling price and CM3 per unit:

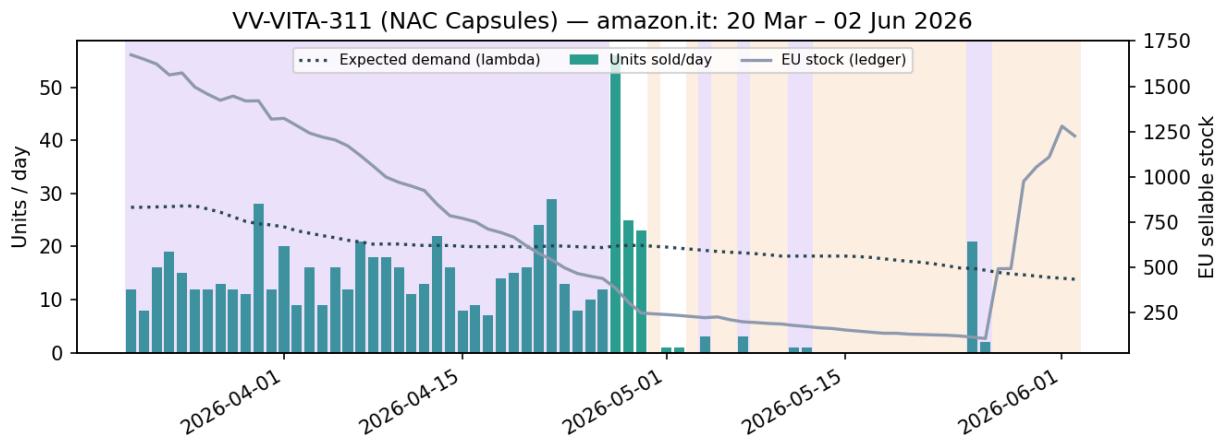
**Lost / Miss revenue = missed units x avg price**

**Lost / Miss CM3 = missed units x avg CM3 per unit** (the true P&L impact).

'Lost' = involuntary (OOS); 'Miss' = voluntary (cooling down). Consecutive flagged days are grouped into discrete events.

## 5. Worked example

**VV-VITA-311 — NAC Capsules**, amazon.it. Full year on this marketplace: **32 OOS days** (lost €10.434 revenue / €1.379 CM3) and **56 cooling-down days** (miss €9.301 revenue / €1.534 CM3).



Highlighted window 20 Mar 2026 – 02 Jun 2026: EU stock ranged from ~1,674 down to ~108 units and reach bottomed at ~6 day(s). In this window the model flagged **26 OOS day(s)** (red/orange shading) and **44 cooling-down day(s)** (purple) — the SKU was throttled (price up / ad spend cut) while stock was tight, then went into genuine stock-out as reach collapsed. Note the balance rarely hits literally zero, so the **reach < 3** rule, not a balance-of-zero rule, is what catches the hard stock-out.

Representative days (Lost/Miss shown as CM3 €):

| Date  | Units | $\lambda$ | EU stk | Reach | Price | Category        | Lost € | Miss € |
|-------|-------|-----------|--------|-------|-------|-----------------|--------|--------|
| 03-20 | 12    | 27        | 1674   | 30    | 19.69 | Cooling down    |        | 43     |
| 03-21 | 8     | 27        | 1650   | 29    | 19.99 | Cooling down    |        | 54     |
| 03-22 | 16    | 27        | 1623   | 29    | 19.88 | Cooling down    |        | 32     |
| 03-23 | 19    | 28        | 1563   | 28    | 19.99 | Cooling down    |        | 24     |
| 03-24 | 15    | 28        | 1574   | 28    | 19.99 | Cooling down    |        | 35     |
| 03-25 | 12    | 28        | 1496   | 27    | 19.99 | Cooling down    |        | 44     |
| 03-26 | 12    | 27        | 1457   | 27    | 19.84 | Cooling down    |        | 43     |
| 03-27 | 13    | 26        | 1423   | 26    | 19.99 | Cooling down    |        | 38     |
| 04-27 | 56    | 20        | 385    | 10    | 16.93 | —               |        |        |
| 04-30 | 0     | 20        | 244    | 7     | —     | Marketplace gap | 45     |        |
| 05-03 | 0     | 19        | 229    | 7     | —     | Marketplace gap | 41     |        |
| 05-05 | 0     | 19        | 227    | 7     | —     | Marketplace gap | 39     |        |
| 05-06 | 0     | 19        | 211    | 7     | —     | Marketplace gap | 38     |        |
| 05-08 | 0     | 19        | 196    | 7     | —     | Marketplace gap | 36     |        |
| 05-09 | 0     | 18        | 191    | 7     | —     | Marketplace gap | 35     |        |
| 05-10 | 0     | 18        | 188    | 7     | —     | Marketplace gap | 35     |        |
| 05-13 | 0     | 18        | 167    | 7     | —     | Marketplace gap | 34     |        |
| 05-14 | 0     | 18        | 163    | 7     | —     | Marketplace gap | 34     |        |
| 05-15 | 0     | 18        | 155    | 7     | —     | Marketplace gap | 34     |        |

## 6. Caveats

Impact figures are estimates (assume the SKU would have sold at its recent rate). Ad-spend-cut detection only applies from when Novadata began reporting Advertising Costs (~Feb 2026); the price-hike lever spans the full year. Reach is EU-pooled, so reach-based rules need ledger coverage for the SKU. All thresholds (3/day, reach 3 & 30, +8%, -50%) are tunable in the dashboard.